MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 06
Chapter 13: Surface Areas and Volumes

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 objective questions (1 mark each).
2. Section B has 2 Short Answer questions (2 marks each). Section C has 2 questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (6 marks). Show all steps clearly.
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show all steps and calculations clearly.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each) [4 Marks]

1. The curved surface area of a cylinder is 264 m^2 and its volume is 924 m^3. The ratio of its diameter to its height is: **[1]**

(a) (a) 3 : 7 (b) 7 : 3 (c) 6 : 7 (d) 7 : 6

2. A cylinder with base radius of 8 cm and height of 2 cm is melted to form a cone of height 6 cm. The radius of the cone is: **[1]**

(a) (a) 4 cm (b) 5 cm (c) 6 cm (d) 8 cm

3. If three metallic spheres of radii 6 cm, 8 cm, and 10 cm are melted to form a single solid sphere, the diameter of the resulting sphere is: **[1]**

(a) (a) 12 cm (b) 24 cm (c) 30 cm (d) 36 cm

4. **Assertion (A):** If a solid cylinder, a solid cone, and a solid hemisphere stand on equal circular bases of radius r and have the same height r, the volume of the cylinder is equal to the sum of the volumes of the cone and the hemisphere.
Reason (R): If a cylinder, a cone, and a hemisphere have equal base radius r and height r, their volumes are in the ratio 3 : 1 : 2. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

5. A juice seller served customers using glasses of inner diameter 5 cm and height 10 cm. The bottom had a hemispherical raised portion reducing capacity. Find the apparent and actual capacity of the glass. (Use $\pi = 3.14$) **[2]**

6. A solid right circular cone of radius 1 cm stands on a hemisphere of radius 1 cm. If the height of the cone equals its radius, find the volume of the solid in terms of π. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

7. A gulab jamun contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends of total length 5 cm and diameter 2.8 cm. **[3]**

8. From a solid cuboidal metallic block of dimensions 15 cm × 10 cm × 5 cm, a cylindrical hole of diameter 7 cm is drilled out right through its thickness. Find the surface area of the remaining block. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (6 Marks) [6 Marks]

9. Shanta runs an industry in a shed which is in the shape of a cuboid surmounted by a half-cylinder. The base of the shed is 7 m × 15 m, and the height of the cuboidal portion is 8 m.

(i) Find the volume of air that the empty shed can hold.

(ii) If the machinery inside occupies a total space of 300 m^3 and there are 20 workers each occupying 0.08 m^3, find the volume of air inside the shed.

[6]

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— **END OF QUESTION PAPER** —

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